



GradeMate

Hybrid ML + DL Automated Short-Answer Grading

Grade short answers with the precision of an examiner.

A M YEESHU (230111) · Yash Goyal (230048)

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Why Grading Is Broken



Manual grading is slow and inconsistent

Hours spent per batch with variable quality



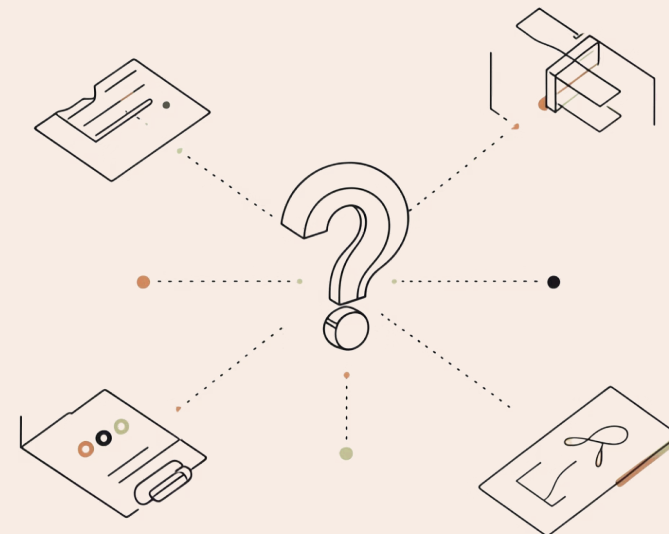
Lexical auto-graders penalise paraphrase

Exact matches rewarded, rephrasing punished



Monolithic deep models hallucinate correctness

Surface form matches mask meaning conflicts



A Stacking Ensemble of ML + DL Learners

We combine TF-IDF, two tree ensembles, and a sentence-transformer regressor through a Ridge meta-model – every prediction is interpretable.



Random Forest

`n_estimators=300, min_samples_split=4`



Gradient Boost

`n_estimators=250, lr=0.05, max_depth=4`



Deep Regressor

MiniLM + MLP architecture



Ridge Meta

Calibrated stacking blend

SciEntsBank Short-Answer Scoring

4,969

Training Rows

Question-answer pairs

540

Test Rows

Held-out evaluation

4

Score Levels

0 to 3, normalised [0, 1]

Q: How do you separate salt from water?

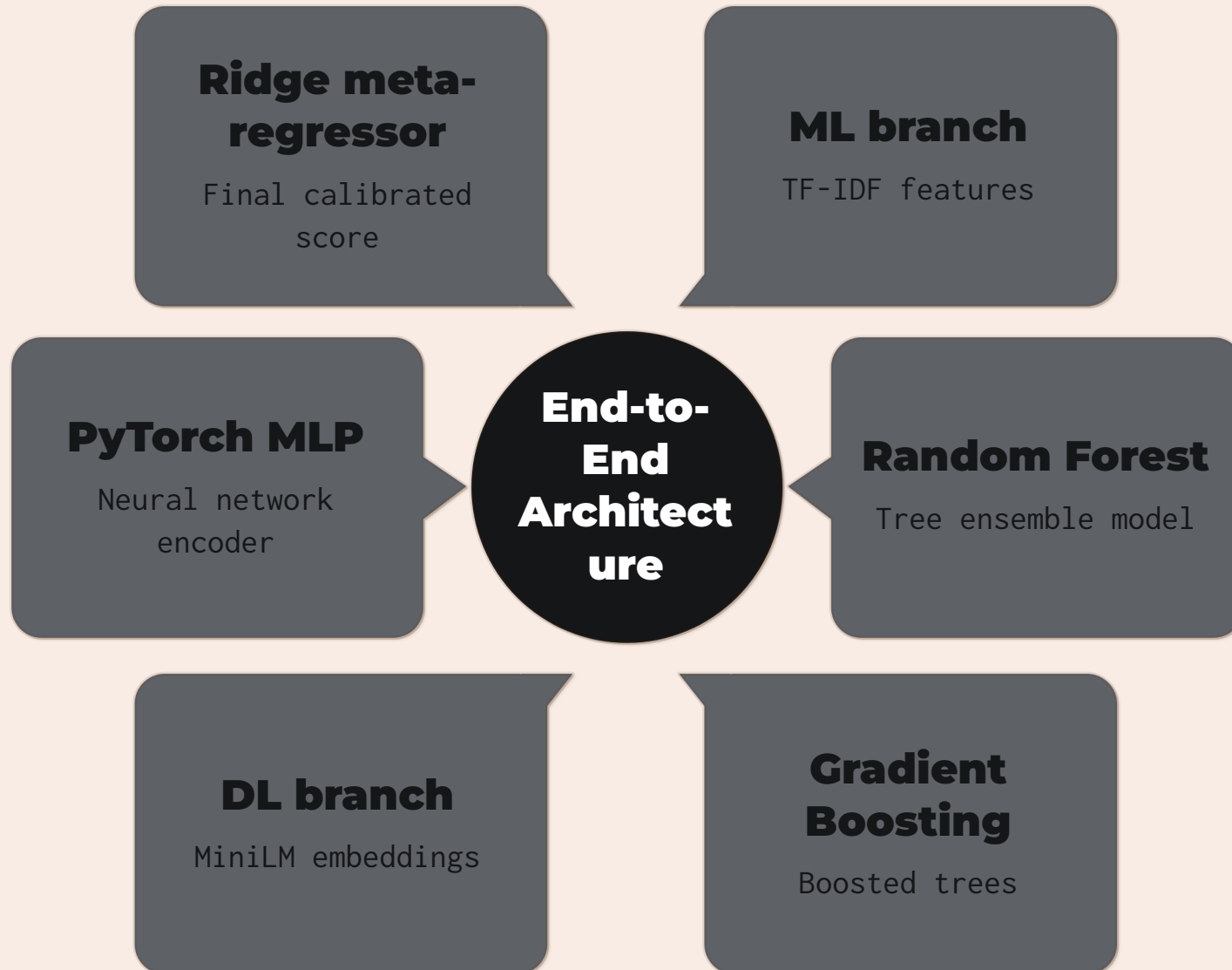
Reference: The water can be evaporated, leaving the salt.

Student: By heating the water until it evaporates.

Label: 0 → target = 1.00

Domain: Science education corpus with expert-scored student responses.

End-to-End Architecture



5-D Engineered Feature Vector

Feature	Description
keyword_overlap	Token Jaccard between cleaned reference and student answers
tfidf_cosine	TF-IDF cosine similarity over (1, 2)-grams
semantic_cosine	L2-normalised MiniLM embedding dot product
length_ratio	$\min(2, \text{student} / \text{ref})$ normalised ratio
bigram_overlap	Fraction of reference bigrams present in student

Lexical + semantic in five numbers.

Classical ML Over the 5-D Vector

Random Forest

- `n_estimators=300`
- `min_samples_split=4`
- `min_samples_leaf=2`

Gradient Boosting

- `n_estimators=250`
- `learning_rate=0.05`
- `max_depth=4`
- `subsample=0.85`

Each model outputs a normalised score in $[0, 1]$.

Deep Regression Head Over MiniLM Embeddings

1536-d Input



The diagram illustrates a deep regression head architecture. It consists of four horizontal layers, each represented by a dark gray arrow pointing to the right. The layers are stacked vertically, with the top layer being the widest and the bottom layer being the narrowest. The text labels for each layer are: '1536-d Input', '256 Neurons', '128 Neurons', and '1 Neuron (Sigmoid)'. The arrows are dark gray, and the text is white.

256 Neurons

128 Neurons

1 Neuron (Sigmoid)

Reading Scanned Answer Sheets

01

OpenCV Pre-process

Deskew → CLAHE → denoise → adaptive threshold

02

EasyOCR / CRAFT

Word-box detection → row grouping (± 25 px y-centre)

03

TrOCR-base-handwritten

Beam-search decoding (num_beams=4, max_new_tokens=128) per line crop

Handwritten answers go through three layers; recognised text feeds the same hybrid grader.

📄 Tesseract kept as a printed-text fallback.